

EDUCATION

Ph.D. Candidate, Computer Science and Engineering Department	Apr 2027 (Expected)
Advisors: Nikhil Bansal, Euiwoong Lee. University of Michigan, Ann Arbor	GPA 3.9/4
Dual Degree (Bachelor and Master of Technology) in Computer Science	May 2022
Indian Institute of Technology (IIT), Kharagpur, India	GPA 9.69/10 Department Rank 5 University Rank 8

RESEARCH INTERESTS

Approximation, Streaming and Parameterized Algorithms for Constraint Satisfiability and Clustering Problems.

PUBLICATIONS (Authors are listed alphabetically by last names in Theory CS.)

Cores Characterize the One-Pass Streaming Complexity of CSPs <i>Joshua Brakensiek, Aaron Putterman, A.S., Santhoshini Velusamy</i>	In submission to ITCS 2026
A Group-based rounding for FPT k-median <i>Aditya Anand, Euiwoong Lee, A.S.</i>	In submission to ITCS 2026
Spectral Dual Fitting for k -Means <i>A. Anand, M. Charikar, V. Cohen-Addad, R. Gao, F. Grandoni, E. Lee, A.S., E. van Wijland</i>	In Submission to SODA 2027
Non-Redundancy of Low-Arity Symmetric Boolean CSPs <i>A.S., Santhoshini Velusamy</i>	In Submission to FSTTCS 2026
Characterizing Streaming Decidability of CSPs via Non-Redundancy <i>A.S., Santhoshini Velusamy</i>	ESA 2026
Min-CSPs on Complete Instances II: Polylog Approximation for Min-NAE-3-SAT <i>Aditya Anand, Euiwoong Lee, Davide Mazzali, A.S.</i>	APPROX 2025
Min-CSPs on Complete Instances <i>Aditya Anand, Euiwoong Lee, A.S.</i>	SODA 2025
A Decomposition Approach to Weighted k-server problem <i>Nikhil Ayyadevara, Ashish Chiplunkar, A.S.</i>	FSTTCS 2024
Two Dimensional Guillotine Strip Packing <i>Arindam Khan, Aditya Lonkar, Arnab Maiti, A.S., Andreas Wiese</i>	ICALP 2022
On Guillotine Separable Packings for the 2D Geometric Knapsack Problem <i>Arindam Khan, Arnab Maiti, A.S., Andreas Wiese</i>	SoCG 2021, HALG 2021
On Parameterized Complexity of Liquid Democracy <i>Palash Dey, Arnab Maiti, A.S.</i>	CALDAM 2021

WORK EXPERIENCE

GSI of EECS 281 Course CSE, UMich	Jan 2026 - Apr 2026
Research Visit Santhoshini Velusamy, TTIC	Sep 2025 - Oct 2025
GSI of EECS 477 Course CSE, UMich	Aug 2024 - Dec 2024
Teaching Assistant Algorithms-1, CSE, IIT Kharagpur	Jan 2022 - Apr 2022
Teaching Assistant Algorithmic Game Theory, CSE, IIT Kharagpur	Aug 2021 - Dec 2021
Research Intern Ashish Chiplunkar, CSE, IIT Delhi	Apr 2021 - Oct 2021
Intern ExaCC, Oracle R&D	May 2021 - Jul 2021
Research Intern Saket Saurabh, University of Bergen	Apr 2020 - May 2020

Research Intern | Arindam Khan, CSA, IISc Bangalore
Research Intern | Arindam Khan, CSA, IISc Bangalore

Dec 2019 - Dec 2019
May 2019 - Jul 2019

SKILLS AND GRAD COURSE WORK

Theoretical CS Approximation and Online Algorithms, Parameterized Algorithms, Randomized Algorithms, Algorithmic Game Theory, Advanced Graph Theory, Computational Geometry, Computational Complexity, Cryptography, Selected Topics in Algorithms.

ACADEMIC ACHIEVEMENTS, FELLOWSHIPS, AND GRANTS

Rackham Conference Travel Funding 2026 <i>Awarded \$1150 in travel support to present at the ESA 2026 Conference.</i>	2026
APPROX 2025 Conference Grant <i>Awarded \$200 in support to present at the APPROX 2025 Conference.</i>	2025
Rackham Conference Travel Funding <i>Awarded \$900 in travel support to present at the APPROX 2025 Conference.</i>	
NSF Travel Support Grant <i>Awarded \$700 in travel support to present at the SODA'25 Conference.</i>	2025
Rackham Conference Travel Funding <i>Awarded \$1400 in travel support to present at the FSTTCS 2024 Conference.</i>	2024
Travel Grant <i>Awarded travel grant for the DIMACS Tutorial on Fine-grained Complexity.</i>	2024
Department Rank <i>Department Rank 5 in the CSE Department, IIT Kharagpur.</i>	2022
University Rank <i>University Rank of 8 at IIT Kharagpur.</i>	2022
GATE Scholarship <i>Awarded for Teaching Assistant-ship at CSE, Kharagpur.</i>	2021
Department Change <i>Moved to CS (in first year) by qualifying in the top 10/1300 at IIT Kharagpur.</i>	2018
JEE Advanced <i>Attained an All India Rank of 1464 among 1.7 lakh students.</i>	2017
SJVN Merit Scholar <i>Awarded for performance in CBSE 12th Examination.</i>	2017
National Talent Search Exam Scholarship <i>All-India top-1000 rank among 600k students and State Rank 1 (Himachal)</i>	2015
Rashtriya Indian Military College Exam <i>Secured National Rank in top 60 in the top Military School in India, State Rank 1.</i>	2013

SERVICE AND EXTRA CURRICULAR

Conference Reviews	SODA'27, SODA'27, Algorithmica'26, RANDOM'26, ICALP'26, ITCS'26, SODA'26, APPROX'24, ISAAC'23, MFCS'23
Theory Talks & Lunches	Organized Theory CS group talks and lunches in the academic year 2023-2024.
Annapurna	An initiative working against hunger and wastage of food resources at IIT Kgp.